

WILLARD NILGES

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EDUCATION

ROCHESTER INSTITUTE OF TECHNOLOGY

May 2023 | GPA: 3.45 - Cum Laude

B.S. Computer Engineering

EXPERIENCE

DATADOG

May 2022 - August 2022 & August 2023 — Present

Software Engineer II - Cognition Platforms

- Building scalable, distributed metadata ingestion, storage, and resolution platforms using **Postgres**, **Cassandra**, **Kafka**, **Redis**, **Python** and **Go**
- Leading team of Cloud Integrations and Cognition Platform engineers to build new metadata storage platform. Enabling faster updates to cloud resources and tags in Datadog's backend. Using **Claude Opus** and **GLM-5** to accelerate development with custom **AI** tools.
- Implemented and migrated to hashring-based load balancing scheme for host intake pipeline. Reduced ingestion latency by 70%, and compute costs of associated ingestion services by 51%.
- Collaborated cross-team to remediate uncontrolled growth of RDS database and prepare colleagues to migrate database safely. Reduced database size by 36% with no downtime.

QUALCOMM

May 2021 — August 2021

Interim Engineering Intern - Core Power

- Implemented debugging suite in **Python**, **C**, and **PRACTICE** to accelerate development for mobile power management subsystems. Reduced time to prepare debugging environment on real hardware from hours to minutes.

COUNCIL ROCK

May 2020 — December 2020 & August 2021 — December 2021

Software Engineering Co-Op

- Overhauled **C++** and **JavaScript** UX/UI for InverSOL home solar solution to improve stability, correctness, and ease of use. Reduced customer support tickets and time to resolution.
- Re-wrote wireless stack in custom **Yocto Linux** firmware to improve solar unit connectivity.
- Designed hybrid-cloud **Kubernetes**-based embedded management platform to enable new market opportunities.

ACTIVITIES

NYC MESH

Technical Lead

- Volunteer-run non-profit wireless ISP dedicated to providing affordable internet access in the five boroughs.
- Led development and implementation of compute infrastructure and software tools to assist volunteers and improve customer experience.
- Presented on work multiple times, including at Texas Linux Fest 2025.

COMPUTER SCIENCE HOUSE

SysAdmin, Executive Board

- Living-learning community focused on technical projects, knowledge, and skills. Managed computing resources and web services for members to facilitate project development.

PROJECTS

WRAITH ENGINE (C)

PRIVATE REPO

- Fully custom 3D engine in **C** for arena movement shooter. Uses **OpenGL**, **Flecs**, and **Open Dynamics Engine**.
- Features data-driven entity component system, serialized level files for asset import with geometry-based collision and physics.

MESHDB (Python)

GITHUB.COM/NYCMESHNET/MESHDB

- **Django** app with **Postgres** database for tracking installations and equipment. Built on distributed baremetal infrastructure.
- Identified and managed customer requirements to improve data availability and reduce install times. Migrated from legacy system with minimal day-to-day disruption.

INKPATH (C++)

GITHUB.COM/WILLNILGES/INKPATH

- Xournal++ plugin using **OpenCV** for transcribing hand-written notes. Contributed API code to popular open source project.

Updated 26th July 2026